

BLUELINE LOGISTICS OPERATIONS PERFORMANCE ANALYSIS (2022-2025)

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SQL// Power BI// DAX

2. Executive Summary

This report presents a comprehensive analysis of logistics operations data covering customer activity, revenue generation, fleet operations, fuel expenditure, maintenance activities, and safety performance.

The project was developed to evaluate operational efficiency and identify factors affecting profitability, fleet utilization, and operational risk within a logistics environment.

Using SQL and Power BI, the dataset was explored, validated, cleaned, analyzed, and transformed into interactive dashboards that provide business insights across three major operational areas:

1. Revenue & Customer Performance
2. Fleet & Fuel Operations
3. Safety & Risk Management

A significant aspect of the project involved identifying and managing data quality issues present within the dataset. Several inconsistencies were discovered during the validation stage, including differing reporting periods across tables, unmatched operational identifiers, and partially populated performance tables.

Rather than forcing unreliable relationships, a domain-based analytical approach was adopted to preserve analytical accuracy and ensure reliable reporting.

The resulting dashboards provide management with visibility into customer revenue patterns, operational cost drivers, maintenance performance, fleet utilization, and safety risks.

The findings highlight opportunities to improve customer diversification, optimize maintenance planning, reduce operational costs, and strengthen safety management practices.

3. Business Problem / Project Background

The logistics industry operates in a highly competitive environment where profitability depends on the efficient movement of goods, effective fleet management, cost control, and operational safety.

Organizations generate large volumes of operational data daily; however, without structured analysis, this information often remains underutilized.

Management teams require visibility into:

1. Revenue performance
2. Customer concentration
3. Route efficiency
4. Fuel expenditure

5. Fleet maintenance costs
6. Safety risks

Without reliable reporting mechanisms, identifying operational inefficiencies and strategic improvement opportunities becomes difficult.

To address this challenge, this project was developed to analyze logistics operational data and transform raw records into actionable business intelligence.

The project focuses on evaluating revenue generation, operational costs, fleet performance, and safety outcomes using SQL and Power BI.

4. Project Objectives

The main objectives of this project were to:

1. Analyze revenue performance across customers and freight activities.
2. Identify the highest revenue-generating customers and routes.
3. Examine revenue contribution by load type and booking type.
4. Evaluate fleet operational performance through fuel expenditure and maintenance activities.
5. Measure fleet downtime and identify high-cost assets.
6. Assess operational safety performance using incident and claims data.
7. Develop interactive Power BI dashboards to support decision-making.
8. Transform raw logistics data into actionable business insights capable of supporting operational improvements.

5. Dataset Overview

The dataset used for this project is the Synthetic Logistics Operations Database obtained from Kaggle.

The database contains multiple interconnected tables representing key logistics business functions including customer management, freight movement, fleet operations, maintenance activities, fuel transactions, driver management, and safety monitoring.

The dataset contains operational records spanning multiple reporting periods and provides sufficient information for evaluating revenue performance, operational efficiency, fleet management, and safety outcomes.

Dataset Source

Yogape Rodriguez (2025).

Synthetic Logistics Operations Database (2022–2024).

Kaggle Dataset.

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LOGISTICS OPERATIONS DATABASE

Data Card

Code (12)

Discussion (1)

Suggestions (0)

About Dataset

Kaggle Dataset: Logistics Operations Database (2022-2024)

About this Dataset

What's Inside

A complete operational database from a fictional Class B trucking company spanning three years.

Usability

10.00

License

MIT

Expected update frequency

Never

5.1 Description of Tables Used

Several tables were selected for analysis based on the objectives of the project.

Customers Table

The Customers table contains information about customers including customer type, account status, freight preferences, and revenue potential. This table was primarily used to analyze customer-related revenue performance.

	customer_id	customer_name	customer_type	credit_terms_days	primary_freight_type	account_status	contract_start_date	annual_revenue_potential
▶	CUST00001	Metro Wholesale	Dedicated	60	General	Inactive	2020-02-20	985117
	CUST00002	National Retail	Contract	30	Retail	Active	2021-06-02	4936566
	CUST00003	XYZ Industries	Contract	30	Consumer Goods	Active	2020-09-04	3102814
	CUST00004	American Corp	Contract	15	Food/Beverage	Active	2020-11-29	3948027
	CUST00005	American Distribution	Spot	45	Consumer Goods	Active	2020-02-04	3682564

Loads Table

The Loads table contains shipment information such as load type, revenue generated, booking type, and shipment status. This table served as the primary source for revenue analysis.

	load_id	customer_id	route_id	load_date	load_type	weight_lbs	pieces	revenue	fuel_surcharge	accessorial_charges	load_status	booking_type
▶	LOAD000000001	CUST00183	RTE00019	2022-01-01	Dry Van	19178	13	3045.23	406.72	100	Completed	Spot
	LOAD000000002	CUST00076	RTE00058	2022-01-01	Dry Van	27761	22	1224.48	98.61	0	Completed	Dedicated
	LOAD000000003	CUST00027	RTE00048	2022-01-01	Refrigerated	35594	16	7171.12	792.88	0	Completed	Spot
	LOAD000000004	CUST00088	RTE00013	2022-01-01	Refrigerated	33274	10	1308.2	141.33	50	Completed	Spot
	LOAD000000005	CUST00185	RTE00020	2022-01-01	Dry Van	40257	10	3317.18	738.48	0	Completed	Spot

Routes Table

The Routes table contains route information including origin and destination locations. It was used to analyze route performance and revenue distribution across transportation corridors.

	route_id	origin_city	origin_state	destination_city	destination_state	typical_distance_miles	base_rate_per_mile	fuel_surcharge_rate	typical_transit_days
▶	RTE00001	Atlanta	GA	Chicago	IL	677	1.7	0.19	1
	RTE00002	Atlanta	GA	Miami	FL	697	2.08	0.22	1
	RTE00003	Chicago	IL	Los Angeles	CA	2003	2.55	0.29	3
	RTE00004	Chicago	IL	Portland	OR	2017	1.84	0.19	3
	RTE00005	Chicago	IL	Las Vegas	NV	1749	2.41	0.26	3

Fuel Purchases Table

This table records fuel transactions, including fuel quantity purchased, location, and associated costs. It was used to evaluate operational fuel expenditure.

	fuel_purchase_id	trip_id	truck_id	driver_id	purchase_date	location_city	location_state	gallons	price_per_gallon	total_cost	fuel_card_number
▶	FUEL00000001	TRIP00051284	TRK00045	DRV00102	2023-10-22 05:00:00	Columbus	MN	131.6	3.399	447.31	FC567161
	FUEL00000002	TRIP00073723	TRK00013		2024-08-04 08:00:00	New York	AZ	139.9	3.18	444.88	FC717910
	FUEL00000003	TRIP00018286	TRK00024	DRV00142	2022-08-23 13:00:00	Seattle	NE	189.3	3.804	720.1	FC912816
	FUEL00000004	TRIP00008465	TRK00018	DRV00047	2022-04-20 00:00:00	Memphis	NY	130.7	3.579	467.78	FC776357
	FUEL00000005	TRIP00069700	TRK00054		2024-06-13 09:00:00	Columbus	AZ	185.3	3.422	634.1	FC391024
	FUEL00000006	TRIP00062650	TRK00012	DRV00104	2024-03-18 23:00:00	Dallas	NY	182.2	3.494	636.61	FC277504
	FUEL00000007	TRIP00000851	TRK00072	DRV00127	2022-01-11 21:00:00	Miami	TX	146.3	3.914	572.62	FC256663
	FUEL00000008	TRIP00012033	TRK00073	DRV00084	2022-06-06 00:00:00	Phoenix	PA	113.3	4.471	506.56	FC543608
	FUEL00000009	TRIP00023596	TRK00101	DRV00080	2022-10-28 02:00:00	Miami	NC	127	4.445	564.51	FC484857

Maintenance Records Table

The Maintenance Records table contains information relating to maintenance activities, repair costs, labor costs, and vehicle downtime.

	maintenance_id	truck_id	maintenance_date	maintenance_type	odometer_reading	labor_hours	labor_cost	parts_cost	total_cost	facility_location	downtime_hours	ser
▶	MAINT00000001	TRK00085	2022-01-01	Inspection	400255	7.8	781.42	10.41	791.83	Kansas City	22.2	Eme
	MAINT00000002	TRK00041	2022-01-01	Tire	268041	0.9	80.91	3207.16	3288.07	Seattle	8	Sch
	MAINT00000003	TRK00090	2022-01-01	Preventive	698915	7.2	733.18	1570.57	2303.75	Miami	16.5	Rou
	MAINT00000004	TRK00056	2022-01-02	Inspection	416522	2.8	298.8	67.48	366.28	Kansas City	34.6	Eme
	MAINT00000005	TRK00029	2022-01-02	Repair	179968	4.6	409.08	982.59	1391.67	Omaha	4.4	Eme
	MAINT00000006	TRK00120	2022-01-02	Preventive	680507	5	498.12	3029.54	3527.66	Omaha	43.5	Sch
	MAINT00000007	TRK00017	2022-01-02	Repair	599078	0.9	83.38	1098.9	1182.28	Denver	34.8	Eme
	MAINT00000008	TRK00038	2022-01-03	Tire	605379	7.1	785.79	1483.25	2269.04	Miami	3.9	Sch

Trucks Table

This table contains fleet asset information including truck status, acquisition details, and equipment specifications.

	truck_id	unit_number	make	model_year	vin	acquisition_date	acquisition_mileage	fuel_type	tank_capacity_gallons	status	home_termin
▶	TRK00001	3463	Peterbilt	2016	1VV205190335317039	2017-04-27	18814	Diesel	200	Active	Omaha
	TRK00002	6461	Kenworth	2015	1NV753749606229960	2018-10-04	26795	Diesel	150	Active	Seattle
	TRK00003	2335	Peterbilt	2018	1NT803101860493229	2020-09-17	11795	Diesel	150	Maintenance	Atlanta
	TRK00004	3317	Freightliner	2015	1VV131945381692351	2017-06-17	8875	Diesel	200	Active	Salt Lake City
	TRK00005	6731	Freightliner	2016	1FT822215709701985	2021-10-08	42042	Diesel	250	Maintenance	Miami
	TRK00006	6082	Kenworth	2017	1VV963516965177788	2018-05-12	47794	Diesel	150	Active	Minneapolis
	TRK00007	7421	Volvo	2019	1MV134100489809236	2020-01-29	42328	Diesel	200	Active	Charlotte
	TRK00008	3112	Mack	2015	1MV426140684859204	2015-01-27	27548	Diesel	150	Active	Houston

Safety Incidents Table

The Safety Incidents table contains records of operational incidents, claims costs, fault status, and incident descriptions.

	incident_id	trip_id	truck_id	driver_id	incident_date	incident_type	location_city	location_state	at_fault_flag	injury_flag	vehicle_damage_cos
▶	INC00000001	TRIP00036079	TRK000006	DRV000006	2023-04-09 14:00:00	Moving Violation	Columbus	PA	True	False	12629.26
	INC00000002	TRIP00032462	TRK000084	DRV000006	2023-02-19 11:00:00	Moving Violation	Columbus	NC	False	False	2700.7
	INC00000003	TRIP00067583	TRK00106	DRV00119	2024-05-20 01:00:00	Customer Complaint	Seattle	OK	True	False	24302.32
	INC00000004	TRIP00049128	TRK00085	DRV00134	2023-09-26 03:00:00	Equipment Damage	Kansas City	TN	True	False	4879.13
	INC00000005	TRIP00062884	TRK00014	DRV00133	2024-03-20 14:00:00	DOT Violation	Las Vegas	IN	False	False	14951.4
	INC00000006	TRIP00004361	TRK000007	DRV000005	2022-02-27 18:00:00	Equipment Damage	Oklahoma City	CO	False	True	0
	INC00000007	TRIP00012773	TRK000007	DRV00109	2022-06-13 02:00:00	DOT Violation	Portland	TX	True	False	12589.05
	INC00000008	TRIP00072329	TRK00115	DRV000060	2024-07-17 20:00:00	Accident	Phoenix	TX	False	False	15572.01

Drivers Table

The Drivers table contains driver-related information including employment status, experience, and licensing details.

	driver_id	first_name	last_name	hire_date	termination_date	license_number	license_state	date_of_birth	home_terminal	employment_status	cdl_class	years_experience
▶	DRV000001	Jennifer	Hernandez	2014-10-31		DL673510887	WA	1973-11-07	Denver	Active	A	3
	DRV000002	William	Martin	2020-10-02		DL128955006	GA	1976-11-03	Columbus	Active	A	20
	DRV000003	Charles	Hernandez	2021-09-21		DL523076025	NC	1970-04-06	Salt Lake City	Active	A	19
	DRV000004	Barbara	Brown	2013-09-08		DL735540030	WA	1995-02-06	Denver	Active	A	19
	DRV000005	Mary	Martinez	2018-12-02		DL706011277	AZ	1960-07-15	Chicago	Active	A	12
	DRV000006	Robert	Taylor	2019-04-04		DL811890569	NC	1990-05-02	Portland	Active	A	23
	DRV000007	James	Martinez	2019-07-02	2021-05-17	DL129100607	PA	1974-05-23	Miami	Terminated	A	5
	DRV000008	Barbara	Wilson	2020-06-09	2021-01-23	DL190137896	CO	1978-12-28	Philadelphia	Terminated	A	15

Trips Table

The trips table contains operational trip information including assigned drivers, trucks, trailers, mileage, duration, fuel usage, and trip status. Initially assessed for operational analysis and relationship validation.

	trip_id	load_id	driver_id	truck_id	trailer_id	dispatch_date	actual_distance_miles	actual_duration_hours	fuel_gallons_used	average_mpg	idle_time_hours	trip_status
▶	TRIP00000001	LOAD000000001	DRV00117	TRK00035	TRL00167	2022-01-01	1314	26.2	183.8	7.15	3.5	Completed
	TRIP00000002	LOAD000000002	DRV00141	TRK00108	TRL00082	2022-01-01	515	8.6	93.6	5.5	8.3	Completed
	TRIP00000003	LOAD000000003	DRV00032	TRK00031	TRL00138	2022-01-01	2509	45	339.1	7.4	12	Completed
	TRIP00000004	LOAD000000004	DRV00083	TRK00105	TRL00018	2022-01-01	717	11.1	110.3	6.5	9.6	Completed
	TRIP00000005	LOAD000000005	DRV00044	TRK00076	TRL00054	2022-01-01	2243	35	328.9	6.82	11.6	Completed
	TRIP00000006	LOAD000000006	DRV00120	TRK00062	TRL00007	2022-01-01	1977	39.4	358.8	5.51	6.6	Completed
	TRIP00000007	LOAD000000007	DRV00055	TRK00002	TRL00093	2022-01-01	1350	20.9	230.8	5.85	2.3	Completed
	TRIP00000008	LOAD000000008	DRV00065	TRK00035	TRL00148	2022-01-01	2748	50.5	437.6	6.28	10.5	Completed
	TRIP00000009	LOAD000000009	DRV00129	TRK00014	TRL00083	2022-01-01	3226	51.4	465.5	6.93	4.1	Completed

Truck Utilization Metrics Table

Contains monthly fleet performance metrics such as mileage, revenue, maintenance costs, downtime, and utilization rates. This table was assessed during the project but had limitations that affected its use in final reporting.

	truck_id	month	trips_completed	total_miles	total_revenue	average_mpg	maintenance_events	maintenance_cost	downtime_hours	utilization_rate
▶	TRK000001	2022-01-01	22	39269	84792.02	6.78	2	4380.98	63.1	0.71
	TRK000001	2022-02-01	27	40787	88809.05	6.23	0	0	0	0.964
	TRK000001	2022-03-01	34	57422	120453.71	6.49	0	0	0	1.097
	TRK000001	2022-04-01	20	34428	73531.28	6.5	2	3548.15	71.8	0.667
	TRK000001	2022-05-01	16	20851	40791.53	6.38	1	1969.42	43.9	0.516
	TRK000001	2022-06-01	31	47085	101659.37	6.46	0	0	0	1.033
	TRK000001	2022-07-01	28	38038	84143.99	6.54	0	0	0	0.903
	TRK000001	2022-08-01	27	35902	82253.78	6.3	1	382.57	20.4	0.871
	TRK000001	2022-09-01	24	40898	88581.76	6.56	1	3372.41	46.5	0.8

Driver Monthly Metrics Table

Contains monthly driver performance metrics such as trips completed, revenue generated, fuel consumption, and delivery performance. Validation revealed that the table contained records for only a small subset of drivers, limiting its suitability for complete driver performance analysis.

	driver_id	month	trips_completed	total_miles	total_revenue	average_mpg	total_fuel_gallons	on_time_delivery_rate	average_idle_hours
▶	DRV00001	2022-01-01	26	36620	79141.59	6.61	5574.7	0.385	8.2
	DRV00001	2022-02-01	9	13515	27133.87	6.69	2095.1	0.333	6.8
	DRV00001	2022-03-01	20	30361	62399.62	6.36	4792.2	0.55	7.5
	DRV00001	2022-04-01	19	28127	68161.13	6.61	4251.4	0.474	8.4
	DRV00001	2022-05-01	14	16071	36256.82	6.35	2573.1	0.214	6.8
	DRV00001	2022-06-01	13	17291	39709.91	6.38	2767.8	0.615	5.5
	DRV00001	2022-07-01	21	32772	66855.24	6.41	5198.6	0.476	7.5
	DRV00001	2022-08-01	20	27010	58514.48	6.35	4331.9	0.45	5.8
	DRV00001	2022-09-01	20	30866	72027.99	6.21	4984.9	0.55	5.8

Facilities Table

Contains information about logistics facilities including location, operating hours, and dock capacity. The table was reviewed during exploratory analysis but was not used extensively in dashboard development.

	facility_id	facility_name	facility_type	city	state	latitude	longitude	dock_doors	operating_hours
▶	FAC00001	Houston Distribution Center	Cross-Dock	Houston	TX	29.7604	-95.3698	125	24/7
	FAC00002	Kansas City Hub	Cross-Dock	Kansas City	MO	39.0997	-94.5786	33	7AM-7PM
	FAC00003	Charlotte Distribution Center	Distribution Center	Charlotte	NC	35.2271	-80.8431	138	24/7
	FAC00004	Chicago Distribution Center	Distribution Center	Chicago	IL	41.8781	-87.6298	38	8AM-5PM
	FAC00005	Indianapolis Hub	Distribution Center	Indianapolis	IN	39.7684	-86.1581	56	8AM-5PM
	FAC00006	Atlanta Warehouse	Distribution Center	Atlanta	GA	33.749	-84.388	149	8AM-5PM
	FAC00007	Detroit Warehouse	Terminal	Detroit	MI	42.3314	-83.0458	44	24/7
	FAC00008	Philadelphia Hub	Cross-Dock	Philadelphia	PA	39.9526	-75.1652	110	8AM-5PM
	FAC00009	Nashville Distribution Center	Terminal	Nashville	TN	36.1627	-86.7816	115	6AM-10PM

Delivery Events Table

Contains event-level operational tracking data including scheduled and actual delivery times, detention minutes, and on-time performance indicators. The table was assessed during data exploration and validation.

	event_id	load_id	trip_id	event_type	facility_id	scheduled_datetime	actual_datetime	detention_minutes	on_time_flag	location_city	location_state
▶	EVT00000001	LOAD00000001	TRIP00000001	Pickup	FAC00034	2022-01-01 18:00:00.000000	2022-01-01 20:58:55.918185	0	False	Houston	TX
	EVT00000002	LOAD00000001	TRIP00000001	Delivery	FAC00046	2022-01-02 23:10:55.918185	2022-01-02 21:30:22.142060	230	True	Detroit	MI
	EVT00000003	LOAD00000002	TRIP00000002	Pickup	FAC00015	2022-01-01 18:00:00.000000	2022-01-01 17:37:26.608430	62	True	Kansas City	MO
	EVT00000004	LOAD00000002	TRIP00000002	Delivery	FAC00050	2022-01-02 02:13:26.608430	2022-01-02 05:55:46.238257	67	False	Indianapolis	IN
	EVT00000005	LOAD00000003	TRIP00000003	Pickup	FAC00001	2022-01-01 08:00:00.000000	2022-01-01 07:28:02.169634	83	True	Columbus	OH
	EVT00000006	LOAD00000003	TRIP00000003	Delivery	FAC00022	2022-01-03 04:28:02.169634	2022-01-03 03:05:02.833256	11	True	Portland	OR
	EVT00000007	LOAD00000004	TRIP00000004	Pickup	FAC00029	2022-01-01 17:00:00.000000	2022-01-01 19:32:56.020030	50	False	Phoenix	AZ
	EVT00000008	LOAD00000004	TRIP00000004	Delivery	FAC00037	2022-01-02 06:38:56.020030	2022-01-02 08:41:06.714927	142	False	Denver	CO
	EVT00000009	LOAD00000005	TRIP00000005	Pickup	FAC00046	2022-01-01 08:00:00.000000	2022-01-01 11:51:35.544202	73	False	Houston	TX

Trailers Table

Contains trailer specifications and status information. This table was reviewed during data exploration but was not utilized extensively in the final dashboards.

	trailer_id	trailer_number	trailer_type	length_feet	model_year	vin	acquisition_date	status	current_location
▶	TRL00001	4290	Refrigerated	53	2016	1AV889081755621178	2018-05-11	Active	Kansas City
	TRL00002	8848	Dry Van	53	2018	1BU942062588832828	2018-11-24	Active	Kansas City
	TRL00003	6147	Dry Van	53	2022	1AV669057807416326	2019-01-28	Active	Milwaukee
	TRL00004	3910	Dry Van	53	2024	1AV544000060905105	2018-08-27	Active	New York
	TRL00005	7524	Dry Van	53	2016	1AV170126369813275	2019-11-22	Active	Las Vegas
	TRL00006	6228	Refrigerated	53	2023	1BV247166900716104	2020-04-27	Active	Los Angeles
	TRL00007	5775	Dry Van	53	2015	1AU542428020643035	2016-03-15	Active	Indianapolis
	TRL00008	9044	Refrigerated	53	2020	1CV458098594511128	2019-04-23	Active	Oklahoma City
	TRL00009	4971	Dry Van	53	2020	1BU184087873073667	2021-05-14	Active	Kansas City

Although all fourteen tables were reviewed during the exploratory phase, not all tables were suitable for final analysis. Certain tables contained incomplete records, inconsistent relationships, or limited operational coverage. As a result, the final dashboards primarily utilized validated data from the Customers, Loads, Routes, Fuel Purchases, Maintenance Records, Trucks, Drivers, and Safety Incidents tables while other tables served supporting or validation purposes.

6. Tools and Technologies Used

The successful completion of this project required the use of database management and business intelligence tools capable of handling data preparation, analysis, and visualization tasks.

SQL

SQL served as the primary analytical tool throughout the project. It was used to explore the dataset, validate data quality, identify inconsistencies, clean records, and answer business questions through aggregation and filtering techniques.

SQL queries were also used to investigate relationships between operational tables and generate summarized datasets for reporting purposes.

Power BI

Power BI was used to transform analytical outputs into interactive dashboards and visual reports.

The platform enabled the development of business intelligence dashboards that provided visibility into revenue performance, fleet operations, fuel expenditure, maintenance activities, and safety risks.

Interactive visualizations, filters, and KPI indicators were used to support data exploration and decision-making.

DAX (Data Analysis Expressions)

DAX was utilized within Power BI to create calculated measures and KPI indicators required for dashboard reporting.

Examples of measures developed include Total Revenue, Total Fuel Spend, Total Maintenance Cost, Total Downtime, Total Claims Cost, and Average Revenue metrics.

These measures enabled the creation of dynamic dashboards capable of responding to user interactions and filters.

7. Data Cleaning and Preparation

Before performing any business analysis, a comprehensive data preparation process was carried out to ensure the reliability and accuracy of the dataset.

Data cleaning is an essential stage in the analytics process because inaccurate, incomplete, or inconsistent data can lead to misleading conclusions and poor decision-making.

The preparation process involved data exploration, validation, identification of inconsistencies, and assessment of relationships between tables.

7.1 Initial Data Exploration

The first stage involved examining the structure of each table to understand available attributes, data types, and potential relationships.

Particular attention was given to identifying:

1. Missing values
2. Duplicate records
3. Date fields
4. Primary and foreign key relationships
5. Numerical fields used in analysis

7.2 Data Validation

Several validation checks were performed to assess data quality.

One of the most significant findings involved inconsistencies in reporting periods across operational tables.

The Loads table contained records primarily within the 2022 reporting period, while the Fuel Purchases table contained records extending into later years. This created challenges when attempting to compare revenue and fuel expenditure directly.

	loads_min_date	loads_max_date
▶	2022-01-01	2022-01-02

	fuel_min_date	fuel_max_date
▶	2022-01-03 15:00:00	2025-01-01 14:00:00

Further validation identified unmatched identifiers between selected operational tables. For example, some records within the Fuel Purchases table could not be linked to corresponding trip records.

7.3 Data Cleaning Decisions

Based on the findings from the validation process, several decisions were made to preserve analytical accuracy.

Rather than forcing unreliable relationships between inconsistent datasets, analyses were grouped into separate business domains:

1. Revenue & Customer Analysis
2. Fleet & Fuel Operations Analysis
3. Safety & Risk Analysis

This approach ensured that each dashboard was built using validated and reliable data sources.

Additionally, tables with incomplete population, such as the Driver Monthly Metrics table, were not used as primary analytical sources because they did not represent the full driver population.

These decisions improved the credibility of the analysis and reduced the risk of producing misleading business insights.

8. Data Analysis Using SQL

Following data validation and preparation, SQL was used to perform exploratory and business-focused analysis across the major operational areas of the logistics business. The objective was to uncover patterns, measure performance, and identify potential areas for operational improvement.

The analysis was grouped into three categories:

1. Revenue & Customer Analysis
2. Fleet & Fuel Operations Analysis
3. Safety & Risk Analysis

8.1 Revenue & Customer Analysis

This phase focused on understanding how revenue was generated across customers, freight types, and transportation routes.

The analysis identified top-performing customers, revenue-generating routes, and the contribution of different load and booking types to overall business performance.

Key metrics analyzed included:

1. Total Revenue
2. Total Loads
3. Average Revenue per Load
4. Average Revenue per Customer

5. Revenue by Route
6. Revenue by Customer
7. Revenue by Load Type
8. Revenue by Booking Type

```
-- Top Customers  
SELECT c.customer_name,  
ROUND(SUM(l.revenue),2) revenue  
FROM staging_loads l  
JOIN staging_customers c  
ON l.customer_id = c.customer_id  
GROUP BY c.customer_name  
ORDER BY revenue DESC;
```

	customer_name	revenue
►	Continental Supply Chain	18148.03
	First Group	13285.26
	Premier Industries	12303.83
	Elite Group	11563.97
	Pacific Logistics	11059.47
	Continental Corp	10863.57
	Pacific Supply Chain	8873.01
	ABC Manufacturing	8729.19
	XYZ Wholesale	8179.02
	Premier Manufacturing	7957.68

Top Revenue-Generating Customers

```

SELECT r.origin_city, r.destination_city,
ROUND(SUM(l.revenue),2) revenue
FROM staging_loads l
JOIN staging_routes r
ON l.route_id = r.route_id
GROUP BY r.origin_city, r.destination_city
ORDER BY revenue DESC;

```

	origin_city	destination_city	revenue
►	Charlotte	Seattle	31720.05
	Philadelphia	Seattle	18230.42
	Miami	Seattle	15844.84
	Columbus	Portland	12718.11
	Phoenix	Philadelphia	12234.89
	Houston	Seattle	11407.26
	Las Vegas	Kansas City	10531.37
	Portland	Detroit	10051.91
	Portland	Columbus	9815.37
	Miami	Denver	9288.82

Revenue Distribution Across Routes

8.2 Fleet & Fuel Operations Analysis

Fleet operations were analyzed to evaluate fuel expenditure, maintenance costs, and vehicle downtime.

The objective was to identify operational cost drivers and determine which fleet assets required greater management attention.

The analysis focused on:

1. Total Fuel Spend
2. Fuel Spend by State
3. Maintenance Cost by Truck
4. Truck Downtime
5. Maintenance Activity Distribution

Due to inconsistencies identified during validation, fuel expenditure and maintenance records were analyzed independently to preserve analytical accuracy.

```
SELECT location_state,  
ROUND(SUM(total_cost),2) AS fuel_expenditure  
FROM staging_fuel_purchases  
GROUP BY location_state;
```

	location_state	fuel_expenditure
▶	MN	39937.03
	AZ	35352.38
	NE	32851.17
	NY	36956.6
	TX	70563.08
	PA	39237.93
	NC	36256.65
	MO	43786.49
	TN	68399.05

Fuel Expenditure by State

```

SELECT truck_id,
ROUND(SUM(total_cost),2) maintenance_cost
FROM staging_maintenance_records
GROUP BY truck_id
ORDER BY maintenance_cost DESC;

```

	truck_id	maintenance_cost
►	TRK00034	9454.43
	TRK00029	7085.35
	TRK00075	6556.22
	TRK00098	6443.74
	TRK00102	6350.22
	TRK00091	6249.2
	TRK00006	6221.44
	TRK00027	5919.75
	TRK00090	5828.79

Maintenance Cost Analysis by Truck

8.3 Safety & Risk Analysis

The safety analysis examined operational incidents and associated claims costs.

The objective was to identify recurring safety concerns, evaluate driver-related risk exposure, and assess preventable incident patterns.

Metrics analyzed included:

1. Incident Count
2. Claims Cost
3. Incident Type Distribution
4. At-Fault Incidents
5. Preventable Incidents
6. Claims Cost by Driver

```

SELECT driver_id,
COUNT(*) AS incidents,
ROUND(SUM(claim_amount),2) AS total_claims
FROM staging_safety_incidents
GROUP BY driver_id
ORDER BY total_claims DESC;

```

	driver_id	incidents	total_claims
►	DRV00014	1	64245.72
	DRV00070	2	61179.09
	DRV00100	2	52171.82
	DRV00142	1	50174.86
	DRV00006	3	49239.16
	DRV00146	1	48640.21
	DRV00119	3	48375.58
	DRV00011	2	48287.93
	DRV00012	1	47959.36

Safety Incident Analysis

9. Dashboard Development

After completing SQL analysis, Power BI was used to transform the analytical outputs into interactive dashboards.

Three dashboard pages were developed to reflect the major business domains identified during analysis.

The dashboards were designed to provide management with a concise overview of operational performance while allowing users to interactively explore underlying data.

9.1 Revenue & Customer Analysis Dashboard

This dashboard focuses on customer performance and revenue generation.

It provides visibility into the customers, routes, freight types, and booking categories contributing most significantly to revenue.

Dashboard Components

KPI Cards

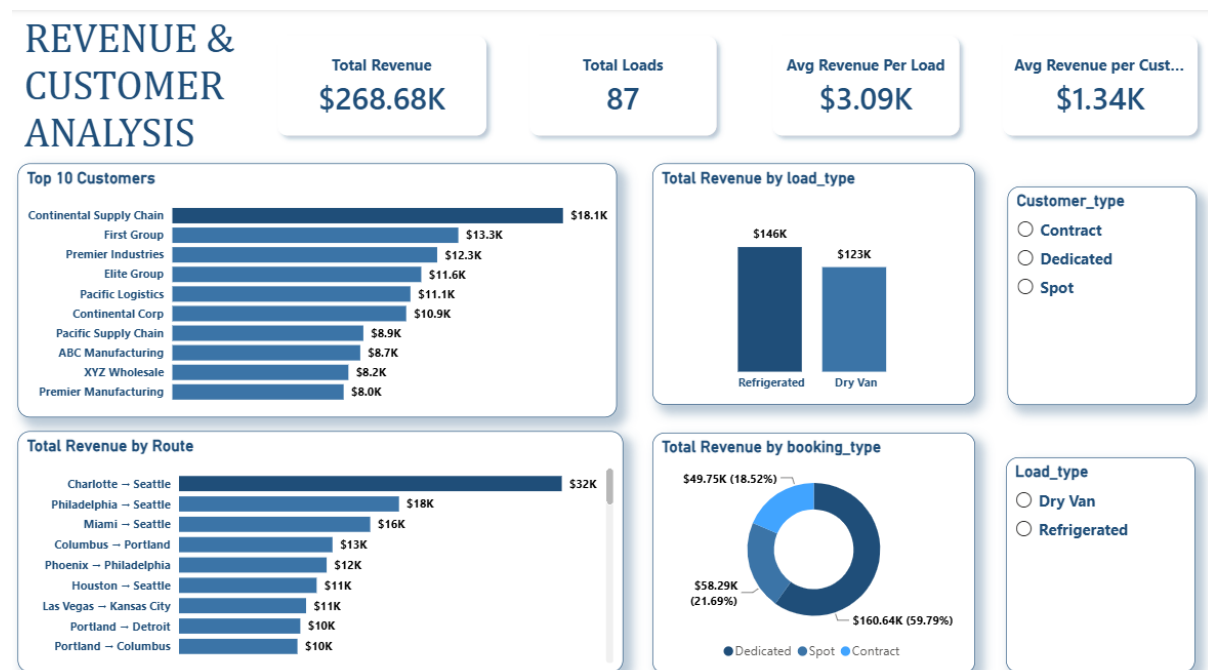
1. Total Revenue
2. Total Loads
3. Average Revenue per Load
4. Average Revenue per Customer

Visualizations

1. Top 10 Customers by Revenue
2. Revenue by Route
3. Revenue by Load Type
4. Revenue by Booking Type

Slicers

1. Customer Type
2. Load Type



Revenue & Customer Analysis Dashboard

9.2 Fleet & Fuel Operations Dashboard

This dashboard provides visibility into operational costs associated with fleet management.

The dashboard enables users to monitor fuel expenditure, maintenance costs, downtime, and vehicle status.

Dashboard Components

KPI Cards

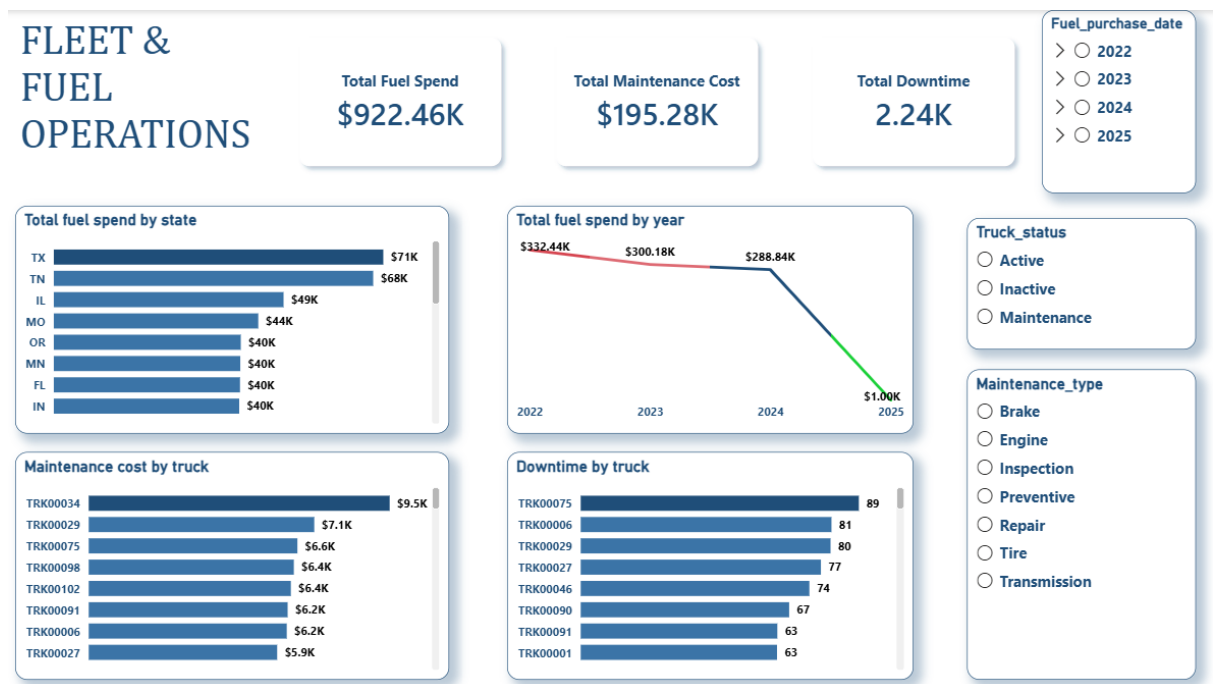
1. Fuel Spend
2. Maintenance Cost
3. Total Downtime

Visualizations

1. Fuel Spend by State
2. Fuel Spend Trend
3. Maintenance Cost by Truck
4. Downtime by Truck

Slicers

1. Maintenance date
2. Fuel purchase date
3. Truck Status



Fleet & Fuel Operations Dashboard

9.3 Safety & Risk Analysis Dashboard

The Safety & Risk dashboard was developed to monitor incident patterns and operational risk factors.

The dashboard enables users to identify high-risk areas, evaluate claims exposure, and monitor driver-related safety performance.

Dashboard Components

KPI Cards

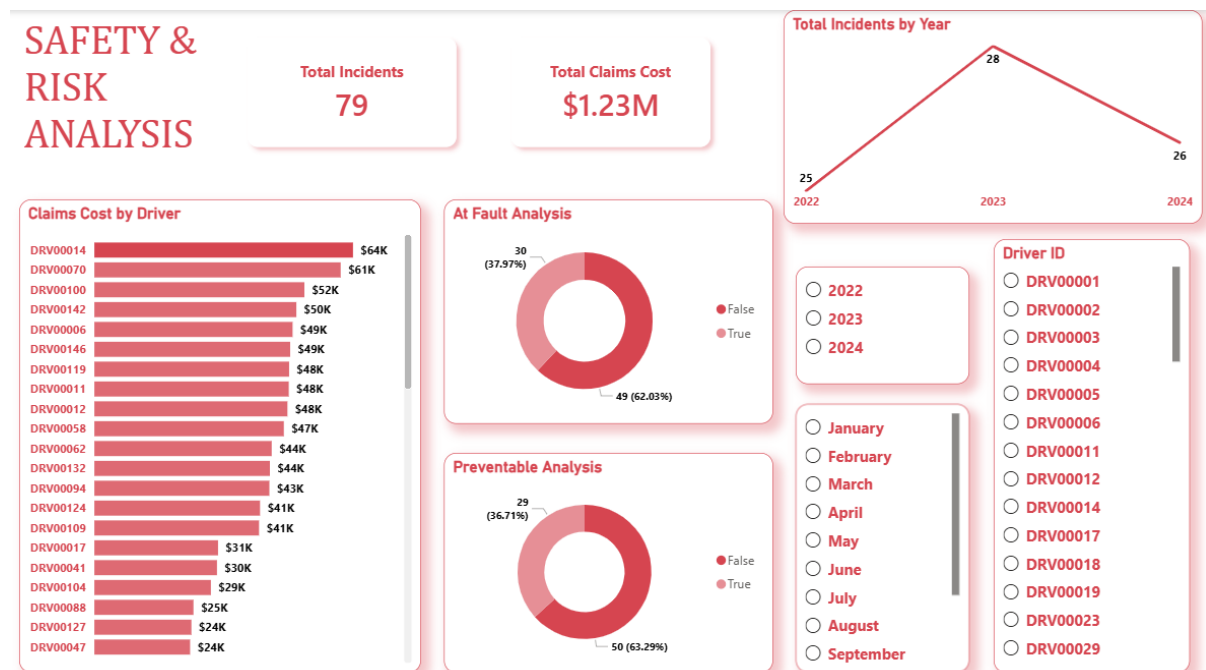
1. Incident Count
2. Claims Cost

Visualizations

1. Incident Types
2. Claims Cost by Driver
3. At-Fault Analysis
4. Preventable Analysis
5. Incident Trend

Slicers

1. Incident Date
2. Incident Type
3. Driver ID



Safety & Risk Dashboard

10. Key Findings

The analysis produced several important insights regarding operational performance.

Revenue generation was concentrated among a relatively small group of customers, indicating a degree of customer dependency within the business. Similarly, certain transportation routes consistently generated higher revenue than others, suggesting that revenue opportunities are concentrated within specific freight corridors.

Fleet analysis revealed that operational costs were not evenly distributed across assets. A limited number of trucks accounted for a significant proportion of maintenance expenses and downtime, highlighting potential opportunities for improved maintenance planning and asset management.

Fuel expenditure patterns also showed geographical concentration, with certain states accounting for higher operational fuel costs than others.

Safety analysis indicated that incidents and claims costs were concentrated among a subset of drivers. Additionally, preventable incidents represented a meaningful portion of total incidents, suggesting opportunities for enhanced safety training and monitoring programs.

11. Recommendations

Based on the findings, the following recommendations are proposed:

Revenue Optimization

1. Reduce dependence on a small number of high-revenue customers by expanding the customer portfolio.
2. Prioritize high-performing freight categories and transportation routes.

Fleet Management

1. Implement more proactive maintenance scheduling for high-cost assets.
2. Monitor vehicles with recurring downtime to improve fleet availability.

Fuel Cost Management

1. Investigate factors contributing to higher fuel expenditure in specific operating regions.
2. Introduce fuel efficiency monitoring initiatives where appropriate.

Safety Improvement

1. Strengthen driver coaching and safety awareness programs.
2. Focus intervention efforts on preventable incident categories to reduce future claims costs.

12. Conclusion

This project demonstrated the practical application of SQL and Power BI in analyzing logistics operations data and generating actionable business insights.

Through structured data validation, business analysis, and dashboard development, the project successfully provided visibility into customer revenue performance, fleet operations, fuel expenditure, maintenance activities, and safety risks.

Despite the data quality challenges encountered during the project, appropriate validation procedures and analytical decisions ensured that findings remained reliable and meaningful.

Overall, the project highlights the importance of data-driven decision-making in logistics management and demonstrates how business intelligence tools can be used to support operational improvement and strategic planning.